Euclid's Elements — Book I
Proposition 17

Formal Reconstruction — Technical Documentation

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Any two angles of a triangle sum to less than two right angles

1 Introduction

Euclid's Proposition I.17 states that in every triangle any two interior angles are together less than two right angles.

For a nondegenerate triangle ABC, the three conclusions are

$$\angle BAC + \angle ABC < 2R,$$

$$\angle BAC + \angle ACB < 2R,$$

and

$$\angle ABC + \angle ACB < 2R.$$

In Euclid's classical proof, Proposition I.16 supplies an exterior-angle inequality and Proposition I.13 identifies an adjacent pair of angles with two right angles. The informal argument then adds the same angle to both sides of an inequality.

The formal reconstruction takes a different route. No numerical angle measure and no binary operation of angle addition are introduced. Instead, the phrase "two angles are together less than two right angles" is represented synthetically: the first angle is required to be strictly smaller than the supplementary angle of the second. This follows the same design principle already used in the reconstruction of Proposition I.14, where "equal to two right angles" was encoded by comparison with a supplementary angle.

The final Lean development consists of a new interface predicate, three pairwise forms of Proposition I.17, and one wrapper theorem combining them.

2 The Classical Configuration

Let ABC be a nondegenerate triangle. Euclid proves a representative case by extending one side of the triangle. In Wilkins's presentation, BC is produced beyond C to a point D:

$$B - C - D.$$

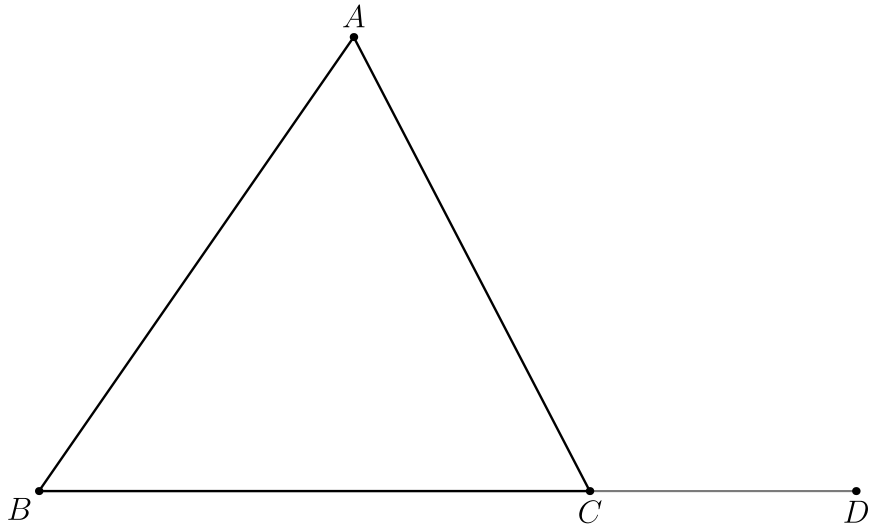

Figure 1: The classical extension used for the pair $\angle ABC, \angle ACB$: BC is produced beyond C.

Proposition I.16 gives

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD.$$

Since CB and CD are opposite rays, $\angle ACB$ and $\angle ACD$ form a supplementary pair. Proposition I.13 therefore gives the classical conclusion

$$\angle ABC + \angle ACB < 2R.$$

The formal reconstruction uses the same extension pattern for two of the three pairwise conclusions. I.16 supplies both

$$\angle BAC < \angle ACD \qquad \text{and} \qquad \angle ABC < \angle ACD,$$

so the geometry $B - C - D$ proves both

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

In the current Lean implementation these two pairwise theorems each call `HilbertOrder.between_extension` independently; the diagram records the common geometric pattern, not identity of their existential witnesses.

For the remaining pair, the formal proof uses the symmetric extension at the other endpoint of the base. Produce CB beyond B to a point E:

$$C - B - E.$$

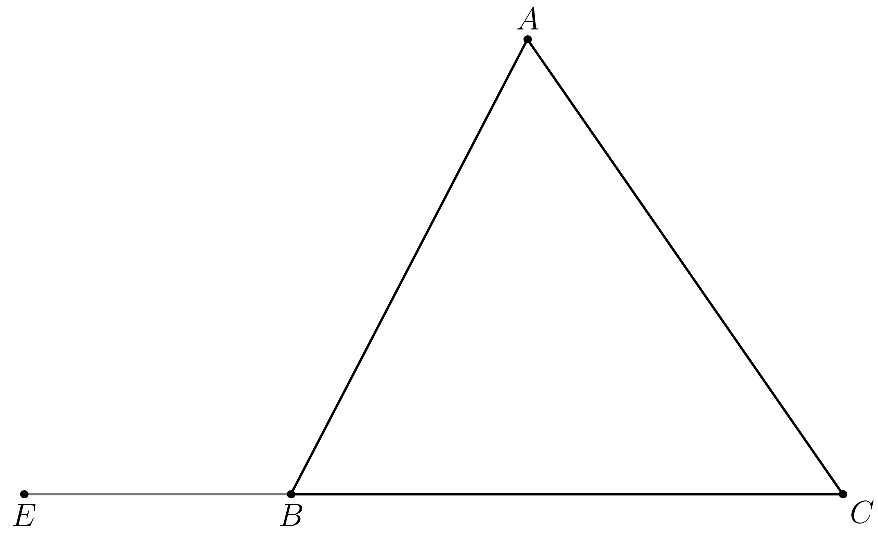

Figure 2: The second extension used by the Lean wrapper: CB is produced beyond B.

Applying I.16 to the reordered triangle ACB yields

$$\angle CAB < \angle ABE.$$

After reversing the first angle,

$$\angle BAC < \angle ABE,$$

and $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

The two figures therefore have genuinely different roles. They are not duplicate views of the same proof: the first extension pattern supplies the two pairs containing $\angle ACB$, while the second supplies the remaining pair.

3 Classical Proof

Euclid proves one representative case and then observes that the same argument applies to the other pairs.

Extend BC beyond C to D. By Proposition I.16,

$$\angle ABC < \angle ACD.$$

Add $\angle ACB$ to both sides:

$$\angle ABC + \angle ACB < \angle ACD + \angle ACB.$$

By Proposition I.13, the adjacent angles $\angle ACD$ and $\angle ACB$ are together equal to two right angles. Hence

$$\angle ABC + \angle ACB < 2R.$$

The other two pairs follow in the same way after choosing the appropriate side extension.

The classical dependency pattern is therefore

$$\begin{array}{c}\text{I.16 exterior-angle inequality}\\ \downarrow\\ \text{add the remaining interior angle}\\ \downarrow\\ \text{I.13 supplementary pair}\\ \downarrow\\ <2R.\end{array}$$

The formal reconstruction deliberately removes the middle arithmetic step.

4 Synthetic Representation of "Less Than Two Right Angles"

The central design choice is the predicate

`HilbertAnglesLessThanTwoRightAngles`.

For two angles α and β, instead of introducing a sum $\alpha + \beta$, choose a supplementary angle β' to β. Then

$$\alpha + \beta < 2R$$

is represented by

$$\alpha < \beta'.$$

In the formal language, if the second angle is $\angle CPD$, a witness E is chosen with

$$D - P - E.$$

Then PE is the opposite ray to PD, so $\angle CPE$ is supplementary to $\angle CPD$. The definition requires

$$\angle AOB < \angle CPE.$$

The Lean definition is:

```
def HilbertAnglesLessThanTwoRightAngles
    [HilbertIncidence Geo]
    (A O B C P D : Geo.Point) : Prop :=
  exists E : Geo.Point,
    Geo.Between D P E /\
    HilbertAngleLess Geo A O B C P E
```

This representation is entirely synthetic. It records only strict betweenness and strict comparison of angles. No real-valued measure, no degree notation, and no algebraic operation on angles is required.

5 Proof Architecture of the Reconstruction

The reconstruction is especially compact because Proposition I.16 already gives exactly the inequalities needed against exterior angles.

For the two pairs containing $\angle ACB$, extend BC beyond C:

$$B - C - D.$$

Then Proposition I.16 gives

$$\angle BAC < \angle ACD$$

and

$$\angle ABC < \angle ACD.$$

Since $\angle ACD$ is the supplement of $\angle ACB$, these are precisely the synthetic forms of

$$\angle BAC + \angle ACB < 2R$$

and

$$\angle ABC + \angle ACB < 2R.$$

For the remaining pair, extend CB beyond B:

$$C - B - E.$$

Apply Proposition I.16 to the reordered triangle ACB. This gives

$$\angle CAB < \angle ABE.$$

Reversal of the first angle identifies $\angle CAB$ with $\angle BAC$, while $\angle ABE$ is supplementary to $\angle ABC$. Hence

$$\angle BAC + \angle ABC < 2R.$$

Schematically:

$$
\begin{array}{c}
\neg\,\text{Collinear}(A,B,C)\\
\downarrow\\
\text{extend } BC \text{ beyond } C\\
\downarrow\\
B-C-D\\
\downarrow\\
\text{I.16}\\
\swarrow \qquad \searrow\\
\begin{array}{cc}
\angle BAC < \angle ACD & \angle ABC < \angle ACD\\
\downarrow & \downarrow\\
\angle BAC + \angle ACB < 2R & \angle ABC + \angle ACB < 2R
\end{array}
\end{array}
$$

and separately

$$
\begin{array}{c}
\text{extend } CB \text{ beyond } B\\
\downarrow\\
C-B-E\\
\downarrow\\
\text{I.16 on triangle } ACB\\
\downarrow\\
\angle CAB < \angle ABE\\
\downarrow\\
\angle BAC + \angle ABC < 2R.
\end{array}
$$

6 Diagrammatic Audit

The current production proof contains two genuinely different incidence/order states and no geometric case split.

- Figure 1 records the extension pattern $B - C - D$. It supports the two pairwise results whose second angle is $\angle ACB$. The two Lean theorems construct their extension witnesses independently, but no additional diagram is justified because their geometry is identical.
- Figure 2 records the distinct extension $C - B - E$ required for the pair $\angle BAC, \angle ABC$ after reordering the triangle.
- The witness X unpacked from `HilbertAngleLess` in the third pair is a representation witness for strict angle comparison. It introduces no new proposition-specific incidence/order state and is therefore not drawn.
- Collinearity symmetry, angle reversal, and the choice between `euclid_proposition_16_first` and `euclid_proposition_16_second` change the logical reading of these configurations, not the configurations themselves.

Thus the final audit retains exactly two figures. A figure records geometry, not theorem-selection or normalization state.

7 Lean Formalization

7.1 The Pair $\angle BAC$ and $\angle ACB$

The first theorem extends BC through C and uses the first half of Proposition I.16.

```
theorem euclid_proposition_17_BAC_ACB
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo B A C A C B := by

  have hBCA :
      not PrimCollinear Geo B C A := by
    intro h

    have hACB :
        PrimCollinear Geo A C B :=
      PrimCollinearSymm Geo B C A h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
```

```
    Geo B C A hBCA

  rcases
      HilbertOrder.between_extension
        B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_first
      Geo A B C D hABC hBCD
```

The witness D simultaneously records the supplementary configuration and serves as the exterior-angle target in I.16.

7.2 The Pair $\angle ABC$ and $\angle ACB$

The second theorem uses the same extension but invokes the second half of I.16.

```
theorem euclid_proposition_17_ABC_ACB
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo A B C A C B := by

  have hBCA :
      not PrimCollinear Geo B C A := by
    intro h

    have hACB :
        PrimCollinear Geo A C B :=
      PrimCollinearSymm Geo B C A h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B hACB

    exact hABC hABC'

  have hBC : B != C :=
    hilbert_noncollinear_ne_first
      Geo B C A hBCA

  rcases
      HilbertOrder.between_extension
        B C hBC with
    <D, hBCD>

  refine <D, hBCD, ?_>

  exact
    euclid_proposition_16_second
      Geo A B C D hABC hBCD
```

Thus the two inequalities associated with the vertex C differ only in which conclusion of I.16 is selected. Each theorem performs its own call to extttbetween_extension; the shared symbol D denotes the same geometric role, not a proof-term-level shared witness.

7.3 The Pair $\angle BAC$ and $\angle ABC$

For the remaining pair, extend CB through B to E and apply I.16 to the triangle in the order ACB.

```
theorem euclid_proposition_17_BAC_ABC
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles
      Geo B A C A B C := by

  have hCBA :
      not PrimCollinear Geo C B A := by
    intro h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearSymm Geo C B A h

    exact hABC hABC'

  have hCB : C != B :=
    hilbert_noncollinear_ne_first
      Geo C B A hCBA

  rcases
      HilbertOrder.between_extension
        C B hCB with
    <E, hCBE>

  refine <E, hCBE, ?_>

  have hACB :
      not PrimCollinear Geo A C B := by
    intro h

    have hABC' :
        PrimCollinear Geo A B C :=
      PrimCollinearRotate Geo A C B h

    exact hABC hABC'

  have hLessCAB :
      HilbertAngleLess Geo C A B A B E :=
    euclid_proposition_16_first
      Geo A C B E hACB hCBE

  rcases hLessCAB with
    <hCABnc, hABEnc, X, hMeet, hAngle>
```

```
have hBACnc :
    not PrimCollinear Geo B A C := by
  intro h
  exact hABC
    (PrimCollinearSwap Geo B A C h)

have hAngle' :
    Geo.AngleCongruent B A C A B X :=
  (Geometry.Geo.angle_congruent_reverse_first
    Geo C A B A B X).mp hAngle

exact
  <hBACnc,
    hABEnc,
    X,
    hMeet,
    hAngle'>
```

The only additional work here is normalization of the orientation of the first angle: I.16 returns $\angle CAB < \angle ABE$, whereas the final interface uses $\angle BAC$. The unordered-angle representation makes this a reversal operation rather than a new geometric argument.

7.4 The Final Wrapper

The final proposition simply combines the three pairwise results.

```
theorem euclid_proposition_17
    [HilbertIncidence Geo]
    [HilbertCongruence Geo]
    (A B C : Geo.Point)
    (hABC : not PrimCollinear Geo A B C) :
    HilbertAnglesLessThanTwoRightAngles Geo B A C A B C /\
    HilbertAnglesLessThanTwoRightAngles Geo B A C A C B /\
    HilbertAnglesLessThanTwoRightAngles Geo A B C A C B := by

  have h1 :=
    euclid_proposition_17_BAC_ABC
      Geo A B C hABC

  have h2 :=
    euclid_proposition_17_BAC_ACB
      Geo A B C hABC

  have h3 :=
    euclid_proposition_17_ABC_ACB
      Geo A B C hABC

  exact <h1, h2, h3>
```

8 Relationship with Propositions I.13, I.14, and I.16

Proposition I.17 sits at an interesting point in the reconstruction because its classical statement appears to require angle arithmetic, while the existing synthetic interface allows that arithmetic to be avoided.

8.1 Proposition I.13

I.13 identifies two adjacent angles on a straight line as a supplementary pair. In the Book Zero reconstruction this is represented structurally by `BookZeroSupplement`, rather than by an equation involving angle sums.

For I.17 we use the same geometric idea but do not need to invoke the theorem I.13 explicitly. The witness produced by `between_extension` already records the opposite ray required to construct the supplementary angle.

8.2 Proposition I.14

I.14 introduced the useful design pattern of representing "equal to two right angles" by congruence with a supplementary angle. The I.17 predicate is the strict-inequality analogue of that pattern:

$$\alpha + \beta = 2R \quad \rightsquigarrow \quad \alpha \cong \mathrm{supp}(\beta),$$

while

$$\alpha + \beta < 2R \quad \rightsquigarrow \quad \alpha < \mathrm{supp}(\beta).$$

Thus I.17 extends an already established interface convention rather than introducing a new angle algebra.

8.3 Proposition I.16

I.16 is the decisive geometric input. Once a side is extended, it provides strict comparison of each remote interior angle with the exterior angle. Since that exterior angle is automatically supplementary to the adjacent interior angle, the conclusions of I.16 are already in the exact form required by `HilbertAnglesLessThanTwoRightAngles`.

This gives the particularly short dependency path

$$\text{betweenness extension} \longrightarrow \text{I.16} \longrightarrow \text{synthetic } < 2R.$$

9 Book Zero and Hilbert Components Used

9.1 `HilbertAngleLess`

The strict comparison relation introduced for I.16 remains the fundamental notion of angle inequality. It is synthetic: one angle is less than another when a congruent copy of the first occurs as a proper interior subangle of the second.

9.2 `HilbertOrder.between_extension`

This construction supplies the points D and E on the required opposite rays:

$$B - C - D \qquad \text{or} \qquad C - B - E.$$

The witness is simultaneously the side extension used by I.16 and the supplementary-ray witness used by the definition of "less than two right angles."

9.3 Collinearity Symmetry and Rotation

The three applications of I.16 require the triangle vertices in different orders. The small uses of `PrimCollinearSymm`, `PrimCollinearRotate`, and `PrimCollinearSwap` normalize noncollinearity hypotheses without adding any geometry.

9.4 Angle Reversal

In the third pair, Proposition I.16 naturally produces the angle $\angle CAB$, while the public form of I.17 uses $\angle BAC$. The theorem `angle_congruent_reverse_first` transfers the congruence across reversal of the first angle.

10 Dependency Summary

The formal dependency structure is

Hilbert incidence + order + congruence
↓
`HilbertAngleLess`
↓
Euclid I.16
↓
`HilbertAnglesLessThanTwoRightAngles`
↓
`I.17_BAC_ABC` `I.17_BAC_ACB` `I.17_ABC_ACB`
↓
`euclid_proposition_17`.

No proposition-local axiom declaration appears in the current source. More importantly, no general arithmetic of angles is introduced merely to reproduce Euclid's phrase "together less than two right angles." The proposition is expressed entirely in the existing synthetic language of rays, betweenness, supplementary configurations, and strict angle comparison.

11 Conclusion

Proposition I.17 is a useful example of the difference between reproducing a classical proof literally and reconstructing its mathematical content over a synthetic interface.

Euclid's paper proof uses an implicit monotonicity principle for angle addition: from $\alpha < \beta$ one passes to $\alpha + \gamma < \beta + \gamma$. A literal formalization would therefore require an angle-sum operation and an order theory for such sums.

The Hilbert/Book Zero reconstruction avoids that additional layer. The statement "two angles are together less than two right angles" is encoded directly by comparison of one angle with the supplement of the other. Once this representation is chosen, Proposition I.16 supplies exactly the needed inequalities, and Proposition I.17 becomes a short structural consequence of side extension and exterior-angle comparison.

The current `Proposition17.lean` source contains no local `sorry` and no local `axiom` declaration. This documentation audit did not rerun `lake build` or `#print axioms`, so no new compilation or axiomatic-status claim is made here.